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Time Varying Beta Risk: An Analysis of Alternative Modelling Techniques

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1. INTRODUCTION

This paper examines the performance of modelling techniques that estimate time varying systematic risk. In equilibrium, rational investors expect an increasing return for any corresponding increase in investment risk, however, this cannot be measured directly giving rise to the need for an empirical approach in estimating beta. In recent years, the general assumption of a stationary risk factor, fundamental to security return models such as the CAPM and the market model, has come under increasing scrutiny and there now exists substantial evidence that systematic risk is unstable.[1]

A number of different techniques have emerged that allow for the modelling and estimation of time varying betas. We focus on the three most commonly used approaches in the literature, namely Bollerslev's (1990) multivariate generalised ARCH model

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(M-GARCH), the time varying heteroskedastic market model suggested by Schwert and Seguin (1990) and time dependent betas estimated with the Kalman Filter algorithm previously used by Black, Fraser and Power (1992) among others.

Bollerslev's (1990) M-GARCH model utilises conditional variance estimates produced by a GARCH(1,1) model to construct a conditional time varying beta series. Several studies have been carried out using the M-GARCH beta including Braun, Nelson and Sunier (1995), McClain, Humphries and Boscan (1996) and Gonzales-Rivera (1996). Although the GARCH(1,1) model is an adequate characterisation of the variation in security returns,[2] other specifications of the conditional variance can also be used. For example, Braun, Nelson and Sunier (1995) adopt an exponential GARCH (EGARCH) specification to formally test for the presence of predictive asymmetry in conditional betas. In addition, threshold ARCH models (TARCH) allow for the incorporation of asymmetric news effects into the estimate of conditional volatility (Glosten, Jagannathan and Runkle, 1993; and Zakoian, 1994).

Schwert and Seguin (1990) augment the simple GARCH methodology by constructing an extended market model that incorporates a single factor model of security return heteroscedasticity. Their specification allows for the generation of a time varying conditional beta series. They found that heteroskedasticity in stock returns is an important issue and may be a cause of the poor performance of beta in cross-sectional tests.[3] Other studies that have used the Schwert and Seguin (1990) model include Koutmos, Lee and Theodossiou (1994) and Episcopos (1996).

The third group of models we examine utilises the Kalman Filter method. This technique recursively estimates time varying betas from an initial set of priors, generating a series of conditional alphas and betas for the market model. The structure of the time varying parameters can be explicitly modelled within this framework to follow any stochastic process. Our tests focus on three forms, namely that betas follow (1) a random walk, (2) a first order autoregressive process with a constant mean and (3) a random coefficient around a constant mean. Each process imposes a different structure to the evolution of conditional betas. When betas follow a random walk, any shock to an asset's

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systematic risk will persist indefinitely into the future. The first order autoregressive with constant mean process assumes that shocks to conditional betas have some persistence but are ultimately mean reverting. This is consistent with the empirical findings in Blume (1971) and other studies. The random coefficient model assumes that a shock in any one period has no effect on future beta values.

Although the approaches to estimating time-varying betas have been applied with some success in the literature, no study has as yet examined them in relation to each other and ultimately in comparison to the simple market model beta. In addition, knowledge of the time varying structure of stochastic betas is uncertain in that there has been no rigorous testing of each method with daily security returns data. Accordingly, our paper aims to provide insights into the way that systematic risk varies through time.

Our results indicate that there is considerable time variation in beta. All the models tested in this paper perform adequately in capturing movements in systematic risk, however, few dominate the simple market model beta under the mean square error criterion when forecasting asset returns. Using in-sample forecasts of industry returns, only the Kalman Filter algorithm, and in particular with the random walk parameterisation, consistently performs better than the simple market model beta. This is surprising given the considerably more complex techniques in estimating GARCH and Schwert and Seguin type betas. In addition, tests of differences between each model's return forecast errors provide strong evidence of independence among the models. This may indicate that the models are capturing different aspects of the time varying characteristics of beta. Finally, all models tend to underestimate the magnitude of asset returns for in-sample forecasts showing a general lack of sensitivity in the models to movements in systematic risk.

In Section 2, we outline the time-varying beta estimation techniques under investigation. A description of our sample and a review of our methodology are provided in Section 3. In Section 4 we present our main tests and conclusions follow in Section 5.

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2. THE MARKET MODEL AND ALTERNATIVE BETA ESTIMATION TECHNIQUES

There is considerable evidence that suggests on both an individual stock level and a portfolio level, market model betas display significant time varying properties. It would therefore appear appropriate to specify time varying risk parameterisations in the form of a conditional beta (β_{it}) in place of a simple (and potentially unrealistic) constant beta. A number of alternative techniques have been suggested in the literature by which β_{it} may be generated.

In this paper we consider three such approaches to the estimation of time-varying betas:

(a) GARCH type models of beta, in particular the multivariate generalised ARCH (M-GARCH) model first introduced by Bollerslev (1990), Nelson's (1991) Exponential GARCH (EGARCH) model and the Threshold ARCH (TARCH) model proposed by Glosten, Jagannathan and Runkle (1993) and Zakoian (1994).
(b) The time varying beta market model suggested by Schwert and Seguin (1990) constructed with the three conditional volatility series obtained from the M-GARCH, EGARCH and TARCH models in (a), and
(c) Time varying beta estimates generated by the Kalman Filter algorithm. We consider three alternative dynamic processes potentially followed by a time dependent beta. The random coefficient (with constant mean) parameterisation imposes the assumption that shocks to the conditional beta have no persistence from period to period. Under a first order autoregressive process with constant mean, shocks to the conditional beta have some persistence but the coefficients eventually return to their mean values. The third structure that we impose is that conditional betas follow a random walk. In this system, shocks to the conditional beta estimates persist indefinitely.

Each estimation technique will now be formally developed beginning with the simple unconditional beta estimated from a market model regression. The market model beta, denoted β^{MM},

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will be used as a benchmark against which other more complex time varying beta estimates will be tested.

(i) The Market Model

An unconditional beta for any asset i may be estimated by a standard Market Model regression, which is defined as:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \tag{1}$$

where:

R_{it} is the return on asset i for period t,
R_{mt} is the return on the market index for period t,
$\beta_i = \text{Cov}(R_{it}, R_{mt})/\sigma^2_{mt}$,
$\alpha_i = \bar{R}_i - \beta_i \bar{R}_{mt}$, and
ε_{it} is the disturbance vector.

The error term, ε_{it}, is assumed to have zero mean and a serially independent and homoscedastic variance-covariance matrix. Under this specification, α_i and β_i are assumed constant over time with β_i representing the systematic risk of asset i.

(ii) GARCH Type Models

(a) The Multivariate Generalised ARCH (M-GARCH) model

In order to compute time varying β's using a GARCH model, it is necessary to calculate time-varying conditional covariance's between the asset and the market as well as the time-varying conditional market variance. The number of coefficients that are required by the GARCH(1,1) specification to be simultaneously estimated is high.[4] However, a common way of reducing the onerous computational burden is to constrain the off-diagonal elements of the past error and variance coefficient matrices to be zero.[5] To obtain the conditional covariance equation, an additional assumption of constant correlation between the returns on the asset and returns on the market is imposed. The conditional variances and covariance are then used to construct a time series of conditional betas, β^G_{it} for any asset i:

$$\beta^G_{it} = \frac{\rho_{im}\sigma_{it}}{\sigma_{mt}}. \tag{2}$$

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(b) The Exponential GARCH Model

The assumption of symmetric responses to negative and positive shocks of equal size fundamental to standard GARCH models has been criticised for several reasons. Black (1976), Christie (1982) and Nelson (1991) have suggested that the stock price volatility reaction to good and bad news may be asymmetric. The arrival of shocks that have a negative impact on asset values causes prices to decline and firms' debt to equity levels to increase. On the other hand, positive shocks increase prices causing a decrease in this leverage effect.

The EGARCH model proposed by Nelson (1991) improves upon the standard GARCH specification by adding additional variables into the conditional volatility equations that differentiates positive shocks from negative shocks. The EGARCH specification constrains the volatility to be non-negative and models the leverage effect to be exponential. The construction of the EGARCH beta series follows in the same fashion as in sub-section 2(ii)(a).

(c) The Threshold ARCH model

The Threshold ARCH model (TARCH) model proposed by Glosten, Jagannathan and Runkle (1993) and Zakoian (1994) is similar to the EGARCH specification in that it attempts to model the asymmetry in the stock price volatility reaction to information shocks by utilising dummy variables to differentiate between positive and negative shocks. The TARCH beta series is constructed as described in sub-section 2(ii)(a).

(iii) Schwert and Seguin Time Varying Beta

The Schwert and Seguin (1990) time varying beta model augments the standard market model by including an additional term that captures the varying level of volatility present in market index returns. This approach can be viewed within the general market model framework, in which beta is time-varying. Specifically, we obtain the Schwert and Seguin beta specification by choosing an additional independent variable to equal the inverse of the conditional volatility of market return series. Accordingly, the model to be estimated becomes:

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$$R_{it} = \alpha_i + b_1 R_{mt} + b_2 \left(\frac{R_{mt}}{\sigma^2_{mt}} \right) + \varepsilon_{it}. \quad (3)$$

This approach imposes a constant mean plus a time-varying term to the conditional beta series. Following Schwert and Seguin (1990), the conditional variance of the market is obtained from a GARCH(1,1) process fitted to the market return series. However, we also fit an EGARCH and Threshold ARCH parameterisation to capture possible asymmetric shock effects in the volatility of asset returns.

(iv) Kalman Filter Approaches to Estimating Beta

A dynamic system such as the time-varying risk return process that has been proposed in this paper can be represented in a general form known as the state space model. In this form we define an observation (or measurement) equation and a transition (or state) equation, which together describe the structure and dynamics of a system. A state space form allows us to incorporate unobserved variables into, and estimate them along with, the observable model to impose a time varying structure to beta. State space models are estimated using a powerful recursive algorithm known as the Kalman Filter (Hamilton, 1994; and Harvey, 1993).

The time varying characterisations of beta that we consider in this paper are:

$$B_{it}^{KAR} = \phi(\beta_{it-1} - \bar{\beta}) + \bar{\beta} + \nu_{it-1} \quad \text{[AR(1)]} \quad (4)$$

$$\beta_{it}^{KRC} = \bar{\beta} + \nu_{it-1} \quad \text{[Random coefficient]} \quad (5)$$

$$\beta_{it}^{KRW} = \beta_{it-1} + \nu_{it-1} \quad \text{[Random walk]} \quad (6)$$

3. DATA AND METHODOLOGY

We use daily returns data on 32 different UK industry sectors for the period 1 January, 1969 to 30 April, 1998.[6] Industry portfolios are useful in this context because of the wide variability in business cycles and risk properties across the firms over these sectors. Thus to the extent that unsystematic risk is reduced through the diversification process (leading to more efficient

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estimation of the beta parameters), changes in business cycles and relative risk of particular industries is maintained. Moreover, the close homogeneity of firms within any sector makes a significant contribution to the definition of our tests. The market index used in this analysis is the FT-All Shares index which consists of approximately 85% by market capitalisation of all firms listed on the London Stock Exchange. The index is value weighted.

Our analysis consists of two parts. Before any tests can be carried out on the performance of methods in estimating time varying beta, the conditional beta series must be modelled, together with an analysis of the distributional properties of the series. The second stage of the analysis involves the determination of the 'best' functional form characterising the conditional beta series for the various industries. We attempt to establish the optimal methodology by constructing in-sample forecasts of the index returns during the sample test period.[7] The accuracy of each forecasted beta series is determined by the estimation of the mean squared error of the forecast where:

$$\mathrm{MSE}_i = \sum (R_{it}^* - R_{it})^2 / T. \tag{7}$$

Tests of significant differences between forecast mean squared error estimates are generated using the modified Diebold and Mariano test statistic as developed by Harvey, Leybourne and Newbold (1997).

4. RESULTS

(i) Distribution of Industry Sector Returns

Table 1 provides information on the distributional properties of the industry sector returns that we use in this study. The mean daily return is approximately 0.05% which corresponds to about 13.8% per annum. The highest returns were concentrated in the banking, telecommunications and pharmaceutical industries (an average of 20% per annum) while several industry sectors experienced average returns as low as 11% per annum.[8] There was no obvious grouping of industry returns over the test period. As would be expected, there is a fairly strong correlation between

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Table 1

Daily Return Summary Statistics and Beta Point Estimates

Industry Sector	*Mean*	*Standard Deviation*	*Skewness*	*Kurtosis*	*Beta Estimate (std. error)*
Banks Merchant (5)	0.0006	0.0135	0.0290	24.14	0.735 (0.013)
Banks Retail (13)	0.0007	0.0133	0.0550	10.76	0.965 (0.011)
Breweries (24)	0.0005	0.0120	0.1348	8.23	0.921 (0.009)
Building (22)	0.0005	0.0121	0.0744	10.06	1.037 (0.008)
Chemical (15)	0.0005	0.0126	−0.1522	8.99	0.965 (0.010)
Distribution (11)	0.0006	0.0106	−0.5982	17.71	0.743 (0.009)
Diversified Industries (8)	0.0005	0.0124	−0.3914	15.55	0.945 (0.010)
Electronics (12)	0.0005	0.0141	−0.0054	7.35	1.073 (0.011)
Engineering (36)	0.0005	0.0103	−0.5054	15.10	0.859 (0.007)
Engine Vehicle Components (8)	0.0004	0.0143	−0.1358	8.23	1.089 (0.011)
Food Production (25)	0.0006	0.0104	−0.2188	12.98	0.874 (0.007)
Household Goods & Textiles (10)	0.0005	0.0132	0.0608	9.27	0.854 (0.012)
Health Care (9)	0.0005	0.0134	0.2086	10.90	0.901 (0.012)
Insurance (12)	0.0005	0.0133	0.0298	8.78	0.997 (0.010)
Investment Trusts (28)	0.0005	0.0090	−0.6287	20.57	0.720 (0.007)
Leisure and Hotels (18)	0.0005	0.0133	0.1528	12.57	1.097 (0.009)
Life Assurance (7)	0.0006	0.0144	−0.0414	12.07	1.079 (0.011)
Media (26)	0.0006	0.0114	−0.7161	21.28	0.877 (0.009)
Oil Exploration & Production (12)	0.0004	0.0172	0.8840	17.46	1.084 (0.024)
Oil Integrated (4)	0.0006	0.0138	−0.0370	6.76	0.911 (0.012)
Mining (6)	0.0004	0.0162	−0.2004	10.72	1.002 (0.015)
Other Financial (19)	0.0005	0.0141	−0.0054	7.35	1.073 (0.011)
Paper and Printing (15)	0.0004	0.0122	−0.0028	12.59	0.869 (0.010)
Pharmaceuticals (12)	0.0007	0.0145	−0.1087	12.55	1.051 (0.012)
Property (26)	0.0005	0.0129	0.0632	13.70	0.959 (0.010)
Retail Foods (10)	0.0006	0.0134	0.0084	8.70	0.995 (0.011)
Retail Goods (28)	0.0005	0.0135	−0.0682	9.77	1.105 (0.009)
Support Services (30)	0.0006	0.0121	−0.3529	15.95	0.876 (0.010)
Telecommunications (9)	0.0007	0.0149	1.2929	25.30	0.714 (0.015)
Tobacco (3)	0.0006	0.0167	0.4667	15.55	1.018 (0.016)
Transport (20)	0.0006	0.0130	0.1045	11.18	0.895 (0.011)
Wines and Spirits (4)	0.0006	0.0144	0.0518	9.01	1.130 (0.011)
Market Index	*0.0005*	*0.0099*	*−2.7992*	*13.50*	

Notes:
This table presents summary statistics for daily returns data of 32 UK industry indices in addition to the FT All Shares index covering the period 1 January, 1969 to 30 April, 1998. The market model generated beta estimates (standard errors in parenthesis) are presented in the final column. The number of firms in each sector are given in parentheses next to the name of the industry.

the total risk of each industry (as measured by standard deviation) and the number of firms in each sector.

For instance, the industry with the largest number of constituent firms (Engineering, 36 companies) has a daily standard deviation of returns of 1.03% whereas the returns on the smallest constituent industry (Tobacco, 3 firms) has a

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standard deviation of 1.67%. Although not a monotonic relationship the correlation between the number of firms in an industry and the standard deviation of returns is high, $\rho = -0.626$, and significantly different from zero at the 1% level of significance, $t = -4.47$. Each industry's return distribution is leptokurtic with kurtosis values reaching a high of 25 and a minimum of 7. As a consequence, every industry sector failed a Jarque-Bera test of normality at the 1% level.

Estimates of the market model beta calculated over the full sample period are provided in Table 1. Point estimate betas range from 1.13 for Wines and Spirits to 0.714 for the telecommunications industry. Although a long estimation period increases the statistical precision of beta estimates, the accuracy of the figure may be quite low. A company or industry's systematic risk can vary quite significantly over time because of changes in operating risk and the business cycles of constituent firms in each industry index.[9] As such, there is strong motivation for the use of the techniques described earlier in estimating beta, especially if the sample period is large and there is doubt about the stability of the firm or industry's systematic risk.

(ii) Modelling Conditional Beta

(a) GARCH Type Models

In this sub-section we model the variance of industry returns according to three alternative ARCH specifications, GARCH(1,1), EGARCH(1,1) and TARCH(1,1). The success of any one model in explaining the conditional volatility of a return series may give valuable insights into the performance of the particular parameterisation in the modelling of conditional beta. Thus, we fit GARCH, EGARCH and TARCH models to the returns data for each industry and the FT All Shares index over the full sample period. Parameter estimates of the GARCH, EGARCH and TARCH models are presented in Tables 2a, 2b and 2c respectively.

An examination of the tables show that all models adequately capture the characteristics of industry return variances. For every industry, the ARCH and GARCH coefficients are significantly different from zero at the 1% level. In addition, with the exception of the Breweries industry, the models are stable with

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Table 2a

Estimate of GARCH(1,1) Model Parameters for Industry Sector Returns

Industry Sector	$c_1 \times 10^{-6}$	a_{11}	b_{11}	$a_{11} + b_{11}$	ρ_{im}
Banks Merchant	7.58*	0.165*	0.809*	0.974	0.53
Banks Retail	4.86*	0.071*	0.901*	0.971	0.71
Breweries	0.04*	0.228*	0.787*	1.015	0.76
Building	3.27*	0.093*	0.883*	0.976	0.84
Chemical	1.44*	0.066*	0.927*	0.992	0.75
Distribution	6.06*	0.125*	0.820*	0.945	0.69
Div. Industries	4.22*	0.077*	0.894*	0.971	0.75
Electronics	1.16*	0.045*	0.950*	0.995	0.75
Engineering	3.53*	0.100*	0.865*	0.965	0.82
Eng. Veh. Comp.	2.78*	0.066*	0.921*	0.987	0.75
Food Production	1.86*	0.076*	0.907*	0.983	0.83
Household/Textiles	7.69*	0.091*	0.867*	0.957	0.64
Health Care	1.40*	0.056*	0.938*	0.994	0.66
Insurance	3.23*	0.057*	0.924*	0.981	0.74
Investment Trusts	7.90*	0.152*	0.745*	0.897	0.78
Leisure and Hotels	2.30*	0.087*	0.901*	0.988	0.81
Life Assurance	9.81*	0.127*	0.826*	0.953	0.74
Media	2.62*	0.103*	0.879*	0.983	0.75
Oil Exploration/Prod.	1.81*	0.076*	0.924*	0.999	0.52
Oil Integrated	1.27*	0.041*	0.953*	0.994	0.65
Mining	9.61*	0.086*	0.874*	0.960	0.61
Other Financial	3.45*	0.151*	0.838*	0.989	0.67
Paper and Printing	3.72*	0.108*	0.869*	0.977	0.70
Pharmaceuticals	3.59*	0.069*	0.915*	0.984	0.71
Property	2.88*	0.117*	0.868*	0.985	0.73
Retail Foods	4.30*	0.076*	0.900*	0.976	0.73
Retail Goods	2.11*	0.086*	0.904*	0.991	0.81
Support Services	2.52*	0.095*	0.890*	0.985	0.71
Telecom.	8.48*	0.090*	0.877*	0.967	0.47
Tobacco	10.0*	0.066*	0.901*	0.967	0.60
Transport	2.84*	0.081*	0.905*	0.986	0.68
Wines and Spirits	3.90*	0.070*	0.911*	0.981	0.77
Market Index	2.86*	0.066*	0.902*	0.968	1.00

Notes:
This table presents GARCH(1,1)parameters for models fitted to each industry return series. The GARCH(1,1) model is defined as:

$$\sigma_{it}^2 = c_1 + a_{11}\varepsilon_{it-1}^2 + b_{11}\sigma_{it-1}^2.$$

The sample period is 1 January, 1969 to 30 April, 1998 and * denotes statistical significance at the 1% level. The correlation coefficient between each industry's return and the FT All Share index return is provided in the last column.

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Table 2b

Estimate of EGARCH(1,1) Model Parameters for Industry Sector Returns

Industry Sector	c_1	δ_i	γ_i	b_{11}
Banks Merchant	−0.709*	0.301*	−0.009*	0.942*
Banks Retail	−0.383*	0.170*	−0.026*	0.971*
Breweries	−0.626*	0.594*	−0.021*	0.980*
Building	−0.364*	0.183*	−0.022*	0.975*
Chemical	−0.230*	0.148*	−0.022*	0.987*
Distribution	−0.771*	0.257*	−0.041*	0.937*
Div. Industries	−0.476*	0.200*	−0.028*	0.964*
Electronics	−0.135*	0.099*	−0.020*	0.993*
Engineering	−0.498*	0.214*	−0.025*	0.964*
Eng. Veh. Comp.	−0.292*	0.159*	−0.023*	0.980*
Food Production	−0.313*	0.170*	−0.025*	0.980*
Household/Textiles	−0.603*	0.206*	−0.010*	0.948*
Health Care	−0.254*	0.158*	−0.005*	0.984*
Insurance	−0.208*	0.113*	−0.022*	0.986*
Investment Trusts	−1.562*	0.295*	−0.062*	0.859*
Leisure and Hotels	−0.271*	0.171*	−0.023*	0.984*
Life Assurance	−0.609*	0.243*	−0.025*	0.951*
Media	−0.439*	0.222*	−0.035*	0.970*
Oil Exploration/Prod.	−0.341*	0.200*	−0.032*	0.977*
Oil Integrated	−0.143*	0.097*	−0.015*	0.992*
Mining	−0.715*	0.173*	−0.024*	0.930*
Other Financial	−0.135*	0.099*	−0.020*	0.993*
Paper and Printing	−0.460*	0.226*	−0.014*	0.967*
Pharmaceuticals	−0.298*	0.156*	−0.009*	0.979*
Property	−0.385*	0.226*	−0.019*	0.976*
Retail Foods	−0.397*	0.172*	−0.024*	0.969*
Retail Goods	−0.282*	0.180*	−0.019*	0.983*
Support Services	−0.397*	0.205*	−0.034*	0.973*
Telecom.	−1.694*	0.276*	−0.017*	0.822*
Tobacco	−0.493*	0.163*	0.002*	0.954*
Transport	−0.371*	0.188*	−0.007*	0.974*
Wines and Spirits	−0.301*	0.151*	−0.025*	0.978*
Market Index	−0.331*	0.181*	−0.039*	0.980*

Notes:
This table presents EGARCH(1,1)parameters for models fitted to each industry return series. The EGARCH(1,1) model is defined as:

$$\ln(\sigma_{it}^2) = c_1 + b_{11}\ln(\sigma_{it-1}^2) + \delta_i\left|\frac{\varepsilon_{it-1}}{\sigma_{it-1}}\right| + \gamma_i\frac{\varepsilon_{it-1}}{\sigma_{it-1}}.$$

The sample period is 1 January, 1969 to 30 April, 1998.
The presence of a leverage effect is tested by the hypothesis that γ (the Leverage Effect Term) is statistically less than 0. The impact is asymmetric if γ is not equal to 0. * Denotes statistical significance at the 1% level.

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Table 2c

Estimate of TARCH(1,1) Model Parameters for Industry Sector Returns

Industry Sector	$c_1 \times 10^{-6}$	a_{11}	γ	b_{11}
Banks Merchant	7.38*	0.154*	0.015*	0.813*
Banks Retail	5.06*	0.055*	0.036*	0.899*
Breweries	0.06*	0.229*	0.076*	0.750*
Building	2.80*	0.067*	0.032*	0.898*
Chemical	1.30*	0.051*	0.023*	0.932*
Distribution	5.83*	0.090*	0.060*	0.829*
Div. Industries	4.05*	0.059*	0.030*	0.899*
Electronics	1.24*	0.035*	0.021*	0.949*
Engineering	3.41*	0.086*	0.023*	0.870*
Eng. Veh. Comp.	2.03*	0.041*	0.027*	0.937*
Food Production	1.80*	0.057*	0.035*	0.911*
Household/Textiles	7.03*	0.077*	0.018*	0.876*
Health Care	1.41*	0.050*	0.011*	0.938*
Insurance	3.24*	0.042*	0.033*	0.924*
Investment Trusts	7.40*	0.124*	0.058*	0.752*
Leisure and Hotels	2.20**	0.065*	0.035*	0.907*
Life Assurance	9.28*	0.104*	0.040*	0.834*
Media	2.74*	0.078*	0.052*	0.880*
Oil Exploration/Prod.	1.22*	0.041*	0.041*	0.940*
Oil Integrated	1.39*	0.036*	0.011*	0.951*
Mining	6.65*	0.069*	0.028*	0.891*
Other Financial	1.24*	0.035*	0.021*	0.949*
Paper and Printing	3.65*	0.096*	0.020*	0.872*
Pharmaceuticals	3.57*	0.066*	0.006*	0.915*
Property	2.64*	0.094*	0.032*	0.878*
Retail Foods	4.63*	0.062*	0.030*	0.898*
Retail Goods	1.96*	0.071*	0.026*	0.909*
Support Services	2.46*	0.073*	0.037*	0.894*
Telecom.	6.66*	0.057*	0.052*	0.895*
Tobacco	10.00*	0.066*	0.001*	0.901*
Transport	2.87*	0.079*	0.005*	0.905*
Wines and Spirits	3.64*	0.048*	0.034*	0.918*
Market Index	1.69*	0.061*	0.052*	0.897*

Notes:
This table presents TARCH(1,1) parameters for models fitted to each industry return series. The TARCH(1,1) model is defined as:

$$\sigma_{it}^2 = c_1 + a_{11}\varepsilon_{it-1}^2 + \gamma\varepsilon_{it-1}^2 D_{t-1} + b_{11}\sigma_{it-1}^2.$$

The sample period is 1 January, 1969 to 30 April, 1998.
If the Leverage Effect Term (γ) is statistically greater than 0 the leverage effect exists. If γ is equal to 0, the news impact is asymmetric. * Denotes statistical significance at the 1% level.

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the conditional coefficients summing to less than 1. Moreover, the leverage effect as measured by the EGARCH and TARCH models (Tables 2b and 2c) is significant for every industry series suggesting that there are asymmetries in the reaction of industry sector returns to good and bad news shocks. The correlations between the industry returns and FT All Shares index are provided in Table 2a. In general, they suggest that there is a strong linear relationship between industry returns and the market portfolio. The highest correlation was 0.84 for the building sector while the lowest correlation was for telecommunications, $\rho_{im} = 0.47$.

The success of ARCH techniques in capturing movements in the conditional variance series justifies the modelling of a conditional beta incorporating these methods. A discussion of the conditional beta estimates generated from the ARCH models will be left to sub-section 4(iii) after the other proposed methods have been analysed.

(b) Schwert and Seguin (1990) Models

To generate the Schwert and Seguin conditional beta series (SS), an estimate of the conditional volatility of the market return is first required. We have developed three models of conditional volatility based upon the GARCH(1,1), EGARCH(1,1) and TARCH(1,1) specifications and these are used to generate three different SS conditional betas. For each industry, an OLS regression was run on the extended market model of Schwert and Seguin (equation (3)) and the resulting SS parameter estimates are provided in Table 3 for the GARCH(1,1) model.[10]

The results show that in general, the additional conditional market volatility measure captures a significant amount of variation in industry portfolio returns. Of the 32 sectors analysed, 26 have statistically significant coefficients for this variable at least at the 5% level. There appears to be no obvious similarity between the industries that have insignificant conditional market volatility coefficients, although the insignificant coefficients for retail goods, retail foods and wine and spirits suggest some form of retail factor affecting the results.

Although most conditional market volatility coefficients are significant, their absolute size is very small in comparison to the

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Table 3

Schwert and Seguin Extended Market Model Coefficients – GARCH(1,1) Process

Industry Sector	$a_1 \times 10^{-5}$	b_1	$b_2 \times 10^{-5}$	R^2
Banks Merchant	23.30*	0.905**	−1.640**	0.29
Banks Retail	16.50*	0.917**	0.461**	0.51
Breweries	5.33	0.970**	−0.477**	0.57
Building	−7.37	1.091**	−0.417**	0.72
Chemical	−6.02	0.944**	0.202*	0.57
Distribution	18.80**	0.810**	−0.653**	0.48
Diversified industries	−2.89	0.950**	−0.053	0.56
Electronics	−8.38	1.014**	0.577**	0.56
Engineering	3.34	0.898**	−0.383**	0.68
Eng. Veh. Comp.	−17.20	1.115**	−0.254	0.56
Food Production	7.57	0.889**	−0.146*	0.68
Household/Textiles	4.19	0.859**	−0.048	0.41
Health Care	1.50	0.966**	−0.623**	0.44
Insurance	−1.73	0.931**	0.644**	0.55
Investment Trusts	7.72	0.793**	−0.703**	0.62
Leisure and Hotels	−8.78	1.192**	−0.923**	0.66
Life Assurance	7.41	1.144**	−0.627**	0.54
Media	13.40*	0.948**	−0.689**	0.57
Oil Exploration/Prod.	−29.90	1.221**	−0.982**	0.27
Oil Integrated	3.70	0.805**	1.030**	0.42
Mining	−15.30	1.065**	−0.614**	0.37
Other Financial	17.50*	1.047**	−1.900**	0.46
Paper and Printing	−4.41	0.963**	−0.917**	0.49
Pharmaceuticals	14.10	0.946**	1.020**	0.51
Property	−0.62	1.119**	−1.550**	0.54
Retail Foods	3.62	0.983**	0.117	0.53
Retail Goods	−6.47	1.107**	−0.0171	0.65
Support Services	15.60*	0.958**	−0.797**	0.51
Telecommunications	24.50	0.572**	1.370**	0.23
Tobacco	0.964	0.867**	1.460**	0.36
Transport	11.1	0.964**	−0.667**	0.46
Wines and Spirits	−6.19	1.141**	−0.101	0.59

Notes:
This table presents Schwert and Seguin extended market model estimated coefficients and OLS regression R^2. The model is defined as:

$$R_{it} = \alpha_i + b_1 R_{mt} + b_2 \left(\frac{R_{mt}}{\sigma^2_{mt}} \right) + \varepsilon_{it}$$

where $\sigma^2_{mt} =$ the conditional volatility of the market index as characterised by a GARCH(1,1) process. The sample period is 1 January, 1969 to 30 April, 1998. * Denotes statistical significance at the 5% level and ** denotes statistical significance at the 1% level. The market index is the FT All Shares index.

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beta coefficient. In each case, the conditional market volatility coefficient is several magnitudes smaller than the simple beta. However, the generation of a conditional beta series involves the scaling of this coefficient by the inverse of the conditional market volatility for each period, *t*. Thus, the additional parameter may still impact upon the measurement of systematic risk.

The explanatory power of the regression equations are high, with a maximum R^2 of 0.72 for the building sector and a minimum of 0.23 for telecommunications. We will delay discussion of the conditional beta series generated from the Schwert and Seguin model until sub-section 4(iii).

(c) Kalman Filter Models

For each transition equation (random coefficient, AR(1), random walk), the Kalman Filter algorithm was applied to every industry sector. Table 4 provides mean conditional beta estimates for the period 1 January, 1969 to 30 April, 1998.

An examination of Table 4 shows that several industry sectors failed to converge to a unique solution within 250 iterations. This is indicative of a misspecification in the transition equation. An analysis of the convergence rates of each model can give a valuable insight into the underlying dynamic process of the conditional beta series. The random walk model appears to give the best characterisation of the time varying betas. Only 1 from 32 industries (Distribution) failed to converge to a unique state space equation. This is in contrast to six industries with the random coefficient model and seven industries with the mean reversion model (AR(1) with constant mean). In addition, the random walk model took a shorter time on average to converge to a solution than any other model. Given that this model appeared to dominate all other models, we follow Coutts, Roberts and Mills (1997) in carrying out a test of parameter constancy for the random walk specification and unsurprisingly find that the null hypothesis of parameter stability is rejected at the 5% level for every industry in the sample.

It is interesting that some models appear to better capture the time varying beta distribution for particular industries than other methods. That is, there is little overlap in the convergence failures for each method. An example is the mining industry

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Table 4

Kalman Filter Mean Model Parameters

Industry Sector	*Random Coefficient Parameterisation*	*AR(1) Parameterisation (Mean Reverting)*	*Random Walk Parameterisation*
	Mean Beta	*Mean Beta*	*State Vector*
Banks Merchant	**FTC**	**FTC**	0.600
Banks Retail	0.981	0.811	1.014
Breweries	0.971	0.889	0.886
Building	1.051	0.979	0.996
Chemical	0.957	0.968	0.992
Distribution	0.705	0.577	**FTC**
Diversified Industries	**FTC**	0.951	0.956
Electronics	**FTC**	1.005	1.121
Engineering	**FTC**	0.869	0.818
Eng. Veh. Comp.	0.923	1.089	1.073
Food Production	0.862	0.851	0.870
Household/Textiles	0.854	0.850	0.855
Health Care	0.682	1.095	0.834
Insurance	1.005	1.020	1.024
Investment Trusts	0.730	0.486	0.657
Leisure and Hotels	**FTC**	0.859	1.024
Life Assurance	1.058	**FTC**	1.046
Media	0.844	**FTC**	0.817
Oil Exploration/Prod.	0.624	**FTC**	1.175
Oil Integrated	0.771	0.901	1.000
Mining	0.809	**FTC**	0.991
Other Financial	1.051	1.005	1.121
Paper and Printing	0.875	0.929	0.797
Pharmaceuticals	1.084	1.502	1.146
Property	0.945	0.913	0.837
Retail Foods	1.007	0.999	1.003
Retail Goods	1.090	1.113	1.121
Support Services	**FTC**	0.933	0.797
Telecommunications	0.637	**FTC**	0.812
Tobacco	1.013	**FTC**	1.150
Transport	0.887	0.524	0.868
Wines and Spirits	1.161	0.968	1.144

Notes:
This table presents the mean Kalman Filter conditional beta estimates for the sample period 1 January, 1969 to 30 April, 1998. For the random coefficient, AR(1) and random walk parameterisations. In turn, the transition equations are defined as:

Random Coefficient: $\beta_{it}^{KRC} = \bar{\beta} + \nu_{it-1}$

AR(1): $\beta_{it}^{KAT} = F(\beta_{it-1} - \bar{\beta}) + \bar{\beta} + \nu_{it-1}$

Random walk: $\beta_{it}^{KRW} = \beta_{it-1} + \nu_{it-1}$.

If the Kalman Filter algorithm fails to converge after 250 iterations, a FTC entry is made (Failed to Converge).

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Table 5

Alternative Beta Estimates

Industry Sector	*Market Model Beta*	*GARCH Betas*			*Schwert & Seguin Betas*			*Kalman Filter Betas*		
		GARCH	*EGARCH*	*TARCH*	*GARCH*	*EGARCH*	*TARCH*	*RC*	*AR*	*RW*
Banks, Merchant	0.735	0.754	0.755	0.768	0.638	0.647	0.637	**FTC**	**FTC**	0.578
Banks, Retail	0.965	1.046	1.051	1.049	0.993	1.000	0.995	0.979	0.951	0.983
Breweries	0.921	0.884	1.021	0.980	0.892	0.895	0.895	0.971	0.889	0.928
Building	1.037	1.061	1.079	1.076	1.006	1.006	1.006	1.068	0.979	1.021
Chemical	0.965	1.023	1.036	1.037	0.977	0.977	0.981	0.960	0.968	1.005
Distribution	0.743	0.783	0.794	0.791	0.703	0.704	0.702	0.706	0.672	**FTC**
Diversified Industries	0.945	1.011	1.027	1.023	0.942	0.939	0.941	**FTC**	0.951	0.956
Electronics	1.073	1.170	1.184	1.186	1.108	1.104	1.110	**FTC**	1.006	1.141
Engineering	0.859	0.891	0.896	0.897	0.835	0.837	0.836	**FTC**	0.869	0.827
Eng. Veh. Comp.	1.089	1.166	1.181	1.183	1.074	1.072	1.076	0.933	1.090	1.094
Food Production	0.874	0.909	0.911	0.910	0.865	0.863	0.865	0.862	0.869	0.870
Household/Textiles	0.854	0.950	0.959	0.959	0.851	0.851	0.855	0.854	0.853	0.863
Health Care	0.901	0.949	0.962	0.964	0.864	0.858	0.865	0.697	1.094	0.848
Insurance	0.997	1.087	1.079	1.075	1.036	1.036	1.036	1.005	1.019	1.029
Investment Trusts	0.720	0.764	0.769	0.748	0.677	0.647	0.637	0.730	0.608	0.656
Leisure and Hotels	1.097	1.097	1.119	1.120	1.041	0.647	0.637	**FTC**	1.082	1.048
Life Assurance	1.079	1.146	1.141	1.139	1.041	1.044	1.043	1.059	**FTC**	1.034
Media	0.877	0.896	0.900	0.900	0.836	0.835	0.833	0.845	**FTC**	0.820

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Oil Exploration	1.084	1.082	1.104	1.112	1.060	1.060	1.061	0.630	**FTC**	1.103
Oil Integrated	0.911	1.000	0.994	0.990	0.974	0.971	0.977	0.789	0.903	0.992
Mining	1.002	1.071	1.100	1.066	0.964	0.968	0.964	0.823	**FTC**	0.995
Other Financial	0.852	0.825	0.822	0.823	0.737	1.104	1.110	0.836	0.765	0.694
Paper and Printing	0.869	0.895	0.913	0.913	0.813	0.817	0.813	0.874	0.856	0.817
Pharmaceuticals	1.051	1.153	1.145	1.143	1.113	1.114	1.117	1.083	1.182	1.125
Property	0.959	0.943	0.959	0.959	0.865	0.865	0.863	0.945	0.889	0.849
Retail Foods	0.995	1.079	1.095	1.091	1.002	1.003	1.002	1.007	0.999	1.017
Retail Goods	1.105	1.151	1.172	1.170	1.104	1.106	1.109	1.091	1.113	1.138
Support Services	0.876	0.889	0.901	0.900	0.828	0.822	0.825	**FTC**	0.932	0.806
Telecommunications	0.714	0.780	0.798	0.774	0.797	0.802	0.801	0.643	**FTC**	0.804
Tobacco	1.018	1.184	1.192	1.197	1.107	1.108	1.111	1.020	**FTC**	1.159
Transport	0.895	0.951	0.954	0.955	0.855	0.856	0.854	0.887	0.774	0.868
Wines & Spirits	1.130	1.199	1.205	1.201	1.124	1.129	1.128	1.161	0.968	1.144

Notes:
This table presents mean conditional beta estimates generated from the models that have been discussed in the text. The estimation period spans 1 January, 1969 to 30 April, 1998. The simple market model beta point estimate is reproduced from Table 1. The Kalman Filter beta representations are random coefficient (RC); autoregressive of order 1 (AR) and random walk (RW). If the Kalman Filter algorithm fails to converge after 250 iterations, a FTC entry will be made (Failed to Converge)

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where the mean reverting AR(1) beta model fails to converge whereas the other two models reach a solution. This is opposite to the engineering industry where the random coefficient model fails to converge and the other two models reach a solution. However, for every industry there is at least one transition model that reaches a unique solution.

(iii) A Comparison of the Conditional Beta Estimates

Table 5 presents a summary of the mean conditional beta estimates over all industries for each method. It can be seen that the conditional beta series have a mean value close to the point estimate market model beta. There is a wide range of mean betas for each industry indicating that the alternative models provide additional information over the market model beta. However, the correlations between the conditional beta estimates (Table 6) show that there are considerable differences in each alternative methods' ability to estimate the conditional beta series. The lower the correlation coefficient the greater the difference between each method. With the exception of the GARCH type conditional beta estimates, there is a striking lack of correlation between each beta series. This is surprising given the success of each technique in capturing the variation of time varying betas (Tables 2–4). The lowest correlation is between the Schwert and Seguin TARCH model and the EGARCH model parameterisations, $\rho = 0.02$, whereas the highest correlation occurs within the GARCH model grouping with a coefficient of approximately 0.96 reflecting the strong similarity between the models.

Further insight on the differences and similarities between the different alternative models can be gained through a graphical analysis of the time series properties of the conditional beta parameters. Accordingly, Figures 1a, 1b and 1c show the time varying beta series for three industries (Engine Vehicle Components, Other Financial and Paper & Printing and Other Financial) in the months surrounding the world markets crash of October 1987. It can clearly be seen that there is considerable variation in each conditional beta series. For each industry, the Kalman Filter beta exhibits most sensitivity to changes in beta whereas the Schwert and Seguin beta displays very little time variation. From each figure, it appears that the GARCH beta is

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Table 6

Average Correlations of Conditional Beta Series

	β^{MG}	β^{EG}	β^{TA}	β^{SSMG}	β^{SSEG}	β^{SSTA}	β^{KFRC}	β^{KFAR}	β^{KFRW}
β^{MG}	1	0.930	0.984	−0.047	−0.039	0.029	0.152	0.060	0.511
β^{EG}	0.930	1	0.945	−0.064	−0.068	0.015	0.145	0.057	0.489
β^{TA}	0.984	0.945	1	−0.045	−0.043	0.026	0.149	0.059	0.494
β^{SSMG}	−0.047	−0.064	−0.045	1	0.894	0.799	0.062	0.036	0.104
β^{SSEG}	−0.039	−0.068	−0.043	0.894	1	0.853	0.062	0.032	0.095
β^{SSTA}	0.029	0.015	0.026	0.799	0.853	1	0.062	0.036	0.091
β^{KFRC}	0.152	0.145	0.149	0.062	0.062	0.062	1	0.557	0.422
β^{KFAR}	0.060	0.057	0.059	0.036	0.032	0.036	0.557	1	0.413
β^{KFRW}	0.511	0.489	0.494	0.104	0.095	0.091	0.422	0.413	1

Notes:
This table presents average correlations between each conditional beta estimation method. The correlations are measured for each industry pair and averaged over the full set of industries to give the overall mean correlation coefficient. The beta representations are β^{MG} = M-GARCH(1,1) beta, β^{EG} = EGARCH beta, β^{TA} = TARCH beta, β^{SSMG} = Schwert and Seguin M-GARCH beta, β^{SSEG} = Schwert and Seguin EGARCH beta, β^{SSTA} = Schwert and Seguin TARCH beta, β^{KFRC} = Kalman Filter random coefficient beta, β^{KFAR} = Kalman Filter AR(1) beta, β^{KFRW} = Kalman Filter random walk beta.

effectively a moving average of the Kalman Filter beta or simply a less sensitive measure of beta variability. In addition, each model exhibits shifts in betas that appear to be independent of the other models.

Although these results are informative and provide a valuable insight into the relative success of each method in capturing the variation in beta through time, they do not formally rank the alternative models in relation to each other. This is left to the next sub-section where the in-sample forecast errors are estimated for each method and compared.

(iv) In-sample Forecast Errors of Alternative Time-varying Beta Estimators

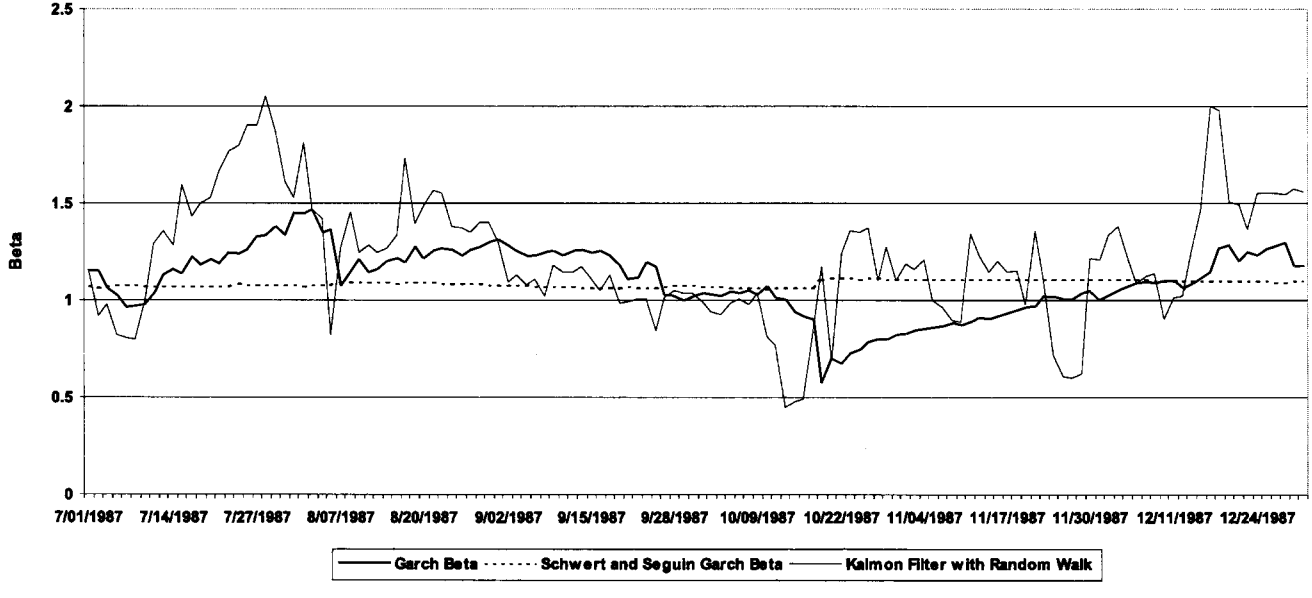
The results from modelling the time-varying conditional beta series of various industries suggest that beta is non-stationary. Moreover, there appears to be significant differences between the alternative methods under investigation. To differentiate between each model, we estimate the in-sample forecast errors of each conditional beta series and determine the best method

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Figure 1a

Engine Vehicle Components Industry Group Beta Series for Last Half of 1987



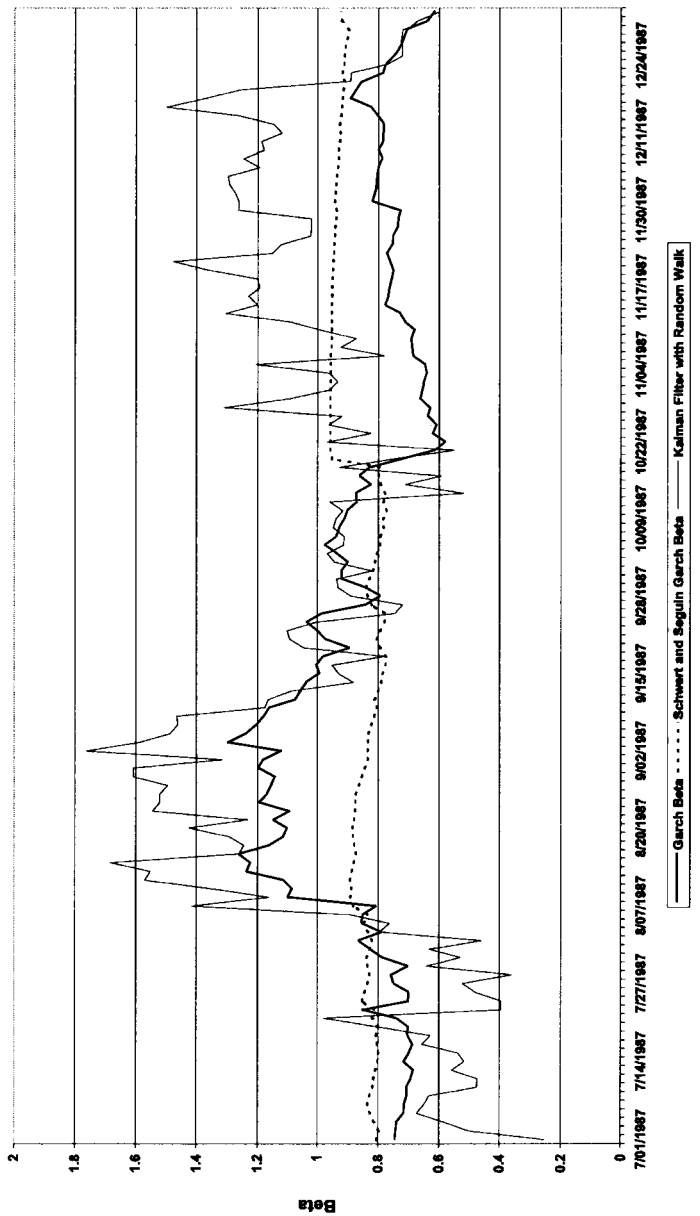
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Figure 1b

Paper and Printing Industry Group Beta Series for Last Half of 1987

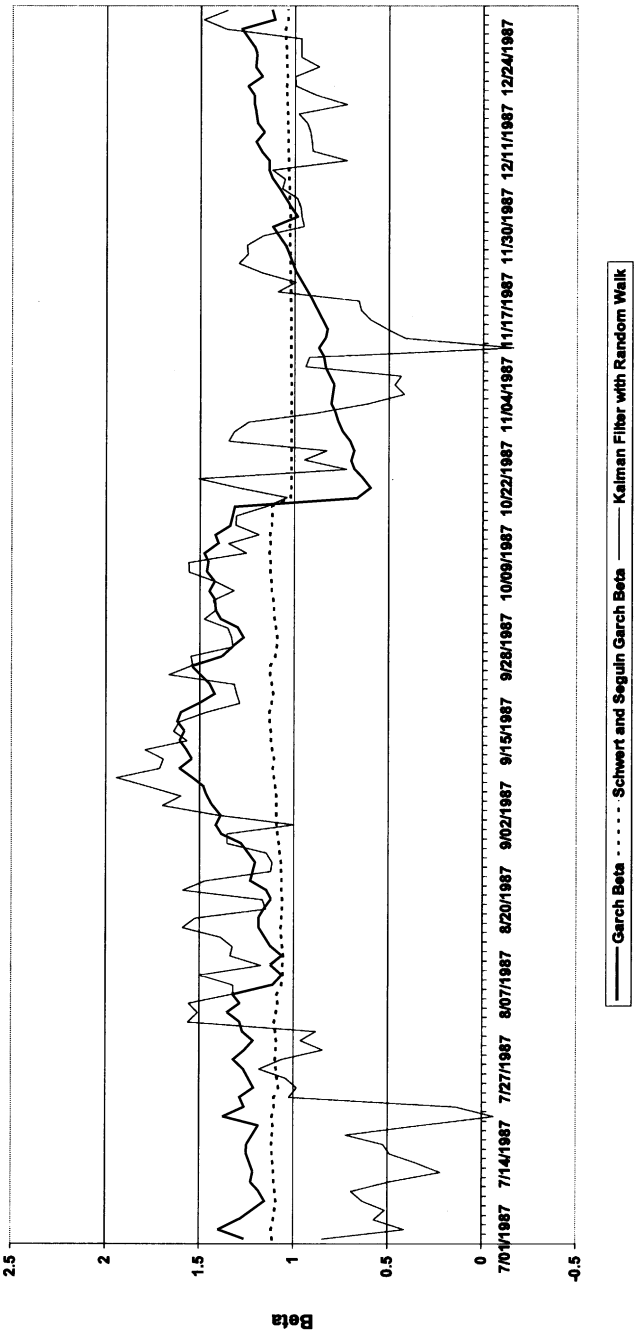


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Figure 1c

Other Financial Industry Group Beta Series for Last Half of 1987



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according to the mean square error (MSE) criterion and using the modified Diebold and Mariano (Harvey, Leybourne and Newbold, 1997) test statistic.

Table 7 presents the mean square errors for each method for each industry sector. It is clear that for the time period under study, the Kalman Filter models dominate with all sectors having at least one of the three Kalman Filter models generating the smallest mean square forecast error. In particular the Kalman Filter with a random walk parameterisation provides the lowest MSE's in 27 of the 32 industry groups. Distribution, oil exploration and production, mining and tobacco are best represented by the Kalman Filter with a random coefficient and only engineering being best modelled by the AR(1) Kalman Filter. The dominance of the Kalman Filter algorithm over all other models is surprising given the significance of the GARCH parameters in capturing the variation in industry volatilities and the success of the Schwert and Seguin extended market models. The GARCH and SS beta estimates generate roughly the same level of mean square error through all industries. There are some exceptions to this result, in particular in those industries where the Kalman Filter random walk method failed to converge. However, for these sectors other Kalman Filter models tended to perform better than the corresponding GARCH and SS betas.

In comparison to the simple market model beta, only the Kalman Filter algorithm consistently generates more efficient forecasts of industry returns. For almost all industries under analysis, market model forecasts are of the same level of efficiency using the mean square error criterion as other complex GARCH and Schwert and Seguin forecasts.

Table 8 provides information on the proportion of industries that rejected the null hypothesis of equal mean square errors by using the modified Diebold and Mariano test statistic. Surprisingly, most conditional beta return forecasts are significantly different from each other. In particular, the Kalman Filter random walk parameterisation leads to different return forecasts from other models in over 90% of the industries tested. However, care should be placed in not over-emphasising the importance of the results in this table since the power of the tests were very high. For each test, more than 7000 forecast observations were compared. This has undoubtedly led to the highly significant test statistics.

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Table 7

Mean Square Errors of In-Sample Forecasts (x 10^{-5})

Industry Sector	β^{MM}	β^{MG}	β^{EG}	β^{TA}	β^{SSMG}	β^{SSEG}	β^{SSTA}	β^{KFRC}	β^{KFAR}	β^{KFRW}
Banks Merchant	13.08	13.16	12.48	13.39	12.59	13.00	12.90	**FTC**	**FTC**	9.11
Banks Retail	8.83	8.99	8.86	8.88	8.84	8.81	8.81	7.95	7.61	6.04
Breweries	6.14	6.37	6.40	5.95	6.18	6.13	6.13	6.12	5.22	4.17
Building	4.36	4.23	4.05	4.05	4.25	4.35	4.35	4.46	4.39	3.39
Chemical	6.90	7.06	6.77	6.75	6.94	6.90	6.90	6.10	5.58	4.77
Distribution	5.87	5.92	5.64	5.64	5.79	5.85	5.84	4.45	4.63	**FTC**
Diversified Industries	6.72	6.65	6.30	6.34	6.41	6.72	6.72	**FTC**	5.45	4.42
Electronics	8.68	8.64	8.26	8.27	8.52	8.66	8.66	**FTC**	8.57	6.04
Engineering	3.40	3.29	3.13	3.16	3.27	3.40	3.40	**FTC**	1.24	1.76
Eng. Veh. Comp.	8.96	8.93	8.49	8.54	8.77	8.96	8.96	7.92	7.74	6.19
Food Production	3.44	3.54	3.39	3.39	3.42	3.44	3.44	3.04	3.41	1.86
Household/ Textiles	10.31	10.44	9.99	9.99	10.15	10.30	10.30	7.87	8.92	7.19
Health Care	10.12	10.36	9.89	9.90	10.16	10.10	10.10	9.97	8.19	6.95
Insurance	6.05	7.93	7.49	7.60	7.79	7.97	7.97	7.13	6.76	5.53
Investment Trusts	3.18	3.36	3.56	3.21	3.21	3.16	3.15	2.21	3.10	2.04
Leisure and Hotels	6.05	6.18	5.84	5.89	5.95	6.00	6.00	**FTC**	5.21	4.14
Life Assurance	9.54	10.15	9.61	9.69	9.51	9.52	9.52	7.25	**FTC**	6.63
Media	5.63	5.82	5.54	5.57	5.70	5.60	5.60	6.01	**FTC**	4.37
Oil Exploration/ Prod.	21.57	22.53	14.46	14.54	22.31	21.50	21.50	9.74	**FTC**	17.44
Oil Integrated	11.11	10.83	10.36	10.33	11.03	11.10	11.10	9.07	9.12	7.66
Mining	16.56	17.05	16.43	16.23	16.54	16.50	16.50	9.78	**FTC**	12.47
Other Financial	8.68	9.03	8.53	8.62	8.72	8.66	8.66	7.64	7.49	5.76
Paper and Printing	7.61	7.91	7.49	7.56	7.56	7.56	7.56	6.01	5.99	5.26
Pharmaceuticals	10.35	10.04	9.63	9.64	9.97	10.30	10.30	7.81	7.95	7.00
Property	7.74	7.81	7.41	7.45	7.75	7.60	7.59	6.77	6.91	5.23
Retail Foods	8.40	8.55	8.14	8.18	8.27	8.40	8.40	7.95	8.08	5.81
Retail Goods	6.36	6.52	6.45	6.24	6.43	6.36	6.36	6.16	5.93	4.85
Support Services	7.14	7.47	7.06	7.13	7.24	7.09	7.10	**FTC**	4.02	4.67
Telecom-munications	17.34	17.39	16.39	16.61	17.55	17.20	17.20	11.62	**FTC**	11.52
Tobacco	18.00	18.17	17.26	17.41	17.76	17.9	17.90	8.19	**FTC**	13.12
Transport	9.15	9.61	9.50	9.65	9.12	9.12	9.12	6.98	7.75	6.94
Wines & Spirits	8.43	8.64	8.60	9.65	8.43	8.43	8.43	7.87	8.68	5.86

Notes:

This table reports mean square error estimates between the observed industry returns series and the in-sample forecast industry returns series. The beta representations are β^{MM} = market model beta, β^{MG} = M-GARCH(1,1) beta, β^{EG} = EGARCH beta, β^{TA} = TARCH beta, β^{SSMG} = Schwert and Seguin M-GARCH beta, β^{SSEG} = Schwert and Seguin EGARCH beta, β^{SSTA} = Schwert and Seguin TARCH beta, β^{KFRC} = Kalman Filter random coefficient beta, β^{KFAR} = Kalman Filter AR(1) beta, β^{KFRW} = Kalman Filter random walk beta.

If the Kalman Filter algorithm fails to converge after 250 iterations, a FTC entry will be made (Failed to Converge)

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Table 8

Percentage of Industries that Reject Null Hypothesis of No Difference in Forecast Error

	β^{MG}	β^{EG}	β^{TA}	β^{SSMG}	*SSEG*	β^{SSTA}	β^{KFRC}	β^{KFAR}	β^{KFRW}
β^{MM}	56.25	46.88	28.13	21.88	18.75	59.38	100	100	100
β^{MG}		56.25	100	18.75	62.50	100	15.63	56.25	90.63
β^{EG}			100	59.38	15.63	100	59.28	3.13	96.88
β^{TA}				100	100	96.88	100	100	96.88
β^{SSMG}					59.38	100	31.25	59.38	90.63
β^{SSEG}						96.88	62.50	21.88	96.88
SSTA							100	100	100
KFRC								59.38	90.63
KFAR									96.88

Notes:
This table presents the proportion of industries that reject the null hypothesis of no difference in MSE forecasts for each conditional beta estimation method. The statistic is the Modified Diebold and Mariano test statistic of Harvey, Leyborne and Newbold (1997). The beta representations are β^{MM} = Market Model beta, β^{MG} = M-GARCH(1,1) beta, β^{EG} = EGARCH beta, β^{TA} = TARCH beta, β^{SSMG} = Schwert and Seguin M-GARCH beta, β^{SSEG} = Schwert and Seguin EGARCH beta, β^{SSTA} = Schwert and Seguin TARCH beta, β^{KFRC} = Kalman Filter random coefficient beta, β^{KFAR} = Kalman Filter AR(1) beta, β^{KFRW} = Kalman Filter random walk beta.

As an additional test, we examine the degree to which forecasts under or over estimate the magnitude of observed returns in the sample period. An underestimate is defined to be any forecast which is less in magnitude than the absolute value of the observed return and an overestimate is any forecast which is greater in magnitude than the observed return. Observed returns and forecasts of opposite sign are treated differently according to the following algorithm: an observed positive(negative) return together with a negative(positive) forecast indicates an under(over)estimate. This allows the determination of the sensitivity of each model forecast to unanticipated changes in observed returns and provides an additional test of the power of each model in capturing the time variation in beta.

Table 9 shows that every method generated forecasts that underestimated the magnitude of the actual returns (the only exception is for the Oil exploration/production industry). Moreover, a sign test with a theoretical percentage of underestimates equal to 50% was significant at the 1% level of significance for each industry/beta model combination. In

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Table 9

Percentage of Underestimated Forecasts for Each Model

Industry Sector	β^{MM}	β^{MG}	β^{EG}	β^{TA}	β^{SSMG}	β^{SSEG}	β^{SSTA}	β^{KFRC}	β^{KFAR}	β^{KFRW}
Banks Merchant	56.63	56.68	56.50	56.60	58.48	58.26	58.49	**FTC**	**FTC**	64.70
Banks Retail	60.27	59.46	59.34	59.47	59.53	59.41	59.54	60.42	61.04	60.87
Breweries	58.06	57.38	52.44	57.98	58.85	58.80	58.78	57.74	58.81	63.79
Building	57.55	57.54	57.55	57.50	58.81	58.96	58.85	57.10	60.27	60.27
Chemical	60.51	59.57	59.79	59.62	60.15	60.12	60.05	61.02	60.47	61.15
Distribution	57.23	56.98	56.98	56.99	58.27	58.23	58.18	58.43	59.74	**FTC**
Diversified Industries	62.40	60.68	60.47	60.62	62.45	62.64	62.52	**FTC**	62.23	62.31
Electronics	60.18	59.54	59.24	59.72	59.71	59.81	59.68	**FTC**	63.33	60.68
Engineering	59.17	58.34	58.42	58.27	60.14	60.04	60.17	**FTC**	58.74	60.14
Eng. Veh. Comp.	60.28	59.20	59.15	59.29	60.78	60.89	60.78	65.13	60.55	61.84
Food Production	57.76	57.07	57.17	57.15	58.15	58.26	58.18	58.15	58.23	58.86
Household/ Textiles	60.38	59.64	59.63	59.57	60.43	60.42	60.34	60.86	60.95	61.97
Health Care	59.68	59.02	59.42	59.20	60.28	60.47	60.25	65.89	54.31	62.13
Insurance	57.20	57.74	57.54	57.65	59.10	59.29	59.19	57.50	57.27	60.68
Investment Trusts	55.44	55.03	54.22	54.89	57.07	56.74	57.08	54.93	61.32	61.46
Leisure and Hotels	57.20	57.74	57.54	57.65	59.10	59.29	59.19	**FTC**	57.27	60.68
Life Assurance	58.34	57.97	58.26	58.08	59.15	59.12	59.12	58.77	**FTC**	60.60
Media	56.23	56.48	56.38	56.29	57.15	57.15	57.21	57.53	**FTC**	60.85
Oil Exploration/ Prod.	43.02	43.67	43.67	43.54	43.14	43.06	43.05	46.56	**FTC**	45.52
Oil Integrated	63.17	61.49	61.71	61.62	61.60	61.80	61.57	67.26	64.52	62.04
Mining	62.52	61.89	61.41	62.22	63.28	63.20	63.30	65.26	**FTC**	63.45
Other Financial	60.18	59.54	59.24	59.72	59.71	59.81	59.68	60.96	63.33	60.68
Paper and Printing	59.20	58.96	58.76	58.85	60.87	60.76	60.90	58.91	59.88	62.28
Pharmaceuticals	58.40	58.02	58.06	58.23	57.08	57.14	57.10	59.15	57.00	58.46
Property	56.36	57.34	57.40	57.50	58.80	59.06	59.04	56.35	57.88	61.81
Retail Foods	59.11	57.74	57.70	57.65	58.91	58.91	58.93	58.80	59.25	62.08
Retail Goods	58.98	57.93	57.81	57.75	59.00	58.93	58.78	59.75	59.00	60.68
Support Services	56.43	56.83	56.66	56.95	57.68	57.79	57.70	**FTC**	56.02	61.84
Telecom-munications	62.02	61.28	60.14	61.23	60.34	60.22	60.19	63.80	**FTC**	63.85
Tobacco	63.85	62.73	62.64	62.68	62.77	62.86	62.82	63.46	**FTC**	64.96
Transport	59.55	59.11	58.82	58.94	60.23	60.25	60.26	60.23	61.25	61.81
Wines & Spirits	60.19	59.04	58.81	58.94	60.39	60.21	60.25	59.30	64.73	60.62

Notes:
This table reports the percentage of under estimated return forecasts for each of the in-sample forecast industry returns series. The beta representations are β^{MM} = market model beta, β^{MG} = M-GARCH(1,1) beta, β^{EG} = EGARCH beta, β^{TA} = TARCH beta, β^{SSMG} = Schwert and Seguin M-GARCH beta, β^{SSEG} = Schwert and Seguin EGARCH beta, β^{SSTA} = Schwert and Seguin TARCH beta, *KFRC* = Kalman Filter random coefficient beta, β^{KFAR} = Kalman Filter AR(1) beta, β^{KFRW} = Kalman Filter random walk beta.
If the Kalman Filter algorithm fails to converge after 250 iterations, a FTC entry will be made (Failed to Converge)

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general the models produced the same underestimation rates for each industry ranging between 55% and 64% for each model. The apparent similarities in underestimation rates and mean square forecast errors (Tables 7 and 8) within each industry is indicative of an omitted industry factor in the models. For example, all the models tended to produce forecasts that overestimated returns for the oil exploration/production industry whereas the same methods tended to underestimate the magnitude of returns for all other industries.

5. CONCLUSIONS

In this paper we examined the ability of alternative models in capturing the time-variation of systematic risk. Our results overwhelmingly suggest that market model betas are unstable and systematically time-varying. Our analysis encompassed three alternative approaches to modelling this time variation in systematic risk: GARCH type models, the Schwert and Seguin extended market model and time-varying betas estimated with the Kalman Filter algorithm. Although all three techniques were successful in characterising the time varying systematic risk of UK industry portfolios, there were also significant differences between each approach. This raises several questions concerning the possibility of unobserved factors (not captured by any one model) affecting the temporal properties of systematic risk. While the simple market model compared favourably with more complex GARCH type models and the Schwert and Seguin model, betas estimated using the Kalman Filter algorithm were consistently more efficient than all other methods. However, our results indicate that a combined method that incorporates the information contained in the conditional volatility of asset returns (as characterised by the various GARCH models) into the Kalman Filter algorithm would be considerably more powerful than any one estimation method in isolation. This is an area worthy of future investigation.

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NOTES

1 See Fabozzi and Francis (1978), Sunder (1980), Bos and Newbold (1984), Collins, Ledolter and Rayburn (1987), Faff, Lee and Fry (1992), Kim (1993), Bos and Fetherston (1995) and Pope and Warrington (1996).
2 See Bollerslev (1986).
3 For example, Chan, Hamao and Lakonishok (1991) and Fama and French (1992, 1993 and 1996).
4 The GARCH(1,1) parameterisation requires 21 individual coefficients and as the order of the GARCH model increases the number of parameters grow exponentially.
5 Pagan (1996) provides a comprehensive discussion of this issue.
6 Our data is taken from Datastream, a financial dataservice.
7 A conditional intercept coefficient α_i is generated by the Kalman Filter process whereas the GARCH and Schwert and Seguin (1990) intercept coefficients are estimated by assuming that α_i is equal to the mean industrial sector return less the mean conditional beta multiplied by the market index.
8 These were engine vehicle components, investment trusts, oil exploration, and paper and printing.
9 We carried out two tests of stationarity (i.e CUSUM, CUSUMSQ) of the market model residuals and found every sector to experience non-stationarity during the test period.
10 The Schwert and Seguin extended market model parameters for the EGARCH and TARCH parameterisations are qualitatively similar to the GARCH(1,1) coefficients. For the purpose of brevity, we have excluded these results from the tables although they are discussed in the text. The parameter estimates for the EGARCH and TARCH models are available from the authors on request.

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